

VideoCD: Contrastive Decoding with Reinforcement Learning for Enhancing Video Temporal Understanding in Video-LLMs

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Abstract

Recent years have witnessed Video-LLM’s rapid progress in video understanding, yet they still lag on temporal reasoning, frequently relying on static cues. Two common remedies exist: 1) Task-specific Supervised Finetuning (SFT) on specially annotated temporal datasets. 2) Adjust internal architectures in LLM. However, SFT is constrained by fixed annotation quality, and modifying model’s architecture consumes abundant workloads when transferred to diverse models. To address these challenges, we propose VideoCD, a novel Reinforcement Learning (RL) framework guided by Contrastive Decoding (CD), which improves Video-LLM’s ability in video understanding through self-reinforcing loop. Aiming to improve Video-LLM’s ability in temporal reasoning, we first construct a video-based Contrastive Decoding scheme, reversing video frames to generate negative samples. Furthermore, we introduce a CD-induced RL Optimization Loop to enhance Video-LLM’s ability permanently, which constructs the reward function based on contrastive signals attained in previous step, continuously guiding the model to iteratively enhance its capability of understanding videos. Throughout the loop, contrastive signals are promoted increasingly and the model is refined by better CD guidance in each loop, effectively solving the annotation quality bottleneck by SFT. Our VideoCD is independent of LLM backbone, thus addressing the resource-consuming challenge of modifying model’s architecture. Experimental results demonstrate significant promotion in model’s accuracy, indicating that model’s temporal reasoning ability has been improved effectively.

1. Introduction

Temporal reasoning stands at the core of video question answering, demanding more than object or action recognition. It requires a coherent understanding of how events unfold, interact, and influence one another across time. A robust model must discern not only what happens, but when and in

Table 1. Performance of various decoding strategies (+VCD, +SID, +TCD) on top of LLaVA-Video across three temporal QA benchmarks.

Dataset	original	+VCD	+SID	+TCD
TempCompass	65.0	65.7	65.8	65.7
Video-MME	59.8	60.0	59.9	59.8
MLVU	67.4	67.3	67.4	67.3

what order, especially in scenarios where causal or sequential dependencies determine the correct answer. Despite significant advances in multimodal foundation models, many still exhibit weak temporal grounding: their responses may be semantically plausible yet inconsistent with the underlying chronology of the video[ReXTime (NeurIPS 2024)V-STaR (CVPR 2025)] .

Current strategies for improving temporal reasoning can be broadly categorized into training time and inference time interventions. On the training side, supervised fine-tuning on temporal QA benchmarks relies heavily on high-quality annotated data [ReXTime, NeurIPS 2024; TemporalBench, 2024]. Model performance is largely bottlenecked by data quality and scale, making further improvements contingent on acquiring even better, more diverse temporal annotations. Architectural modifications such as adding explicit temporal encoders, frame reordering modules, or memory-augmented backbones require model redesign and introduce significant complexity[Chen et al., ICCV 2023; Li et al., ICCV 2023]. Adapting these customized designs to other MLLMs entails considerable engineering effort and hinders portability. On the inference side, contrastive decoding has gained traction as a lightweight plug in[Li et al., 2023] : by comparing model outputs on the original video versus its temporally reversed counterpart, it suppresses time agnostic biases and nudges generation toward temporally coherent responses. While conceptually elegant, this method faces three fundamental limitations. First, it doubles inference latency due to the need for two full forward passes.

Second, its performance gains over the base model are often modest as shown in Table 1, indicating that simple logit subtraction does not fully leverage the contrastive signal. Most importantly, contrastive decoding does not internalize temporal understanding. It acts as a transient correction applied at inference, leaving the model’s internal representations unchanged [Chang et al., EMNLP 2024]. Once the contrast is removed, the model reverts to its original, time insensitive behavior, revealing that no lasting capability has been acquired.

Contrastive decoding operates at inference time, where it compares model outputs on original and temporally altered videos to steer decoding toward temporally coherent answers. And Reinforcement learning leverages output-level evaluations as training-time feedback signals that iteratively refine the model’s policy to internalize desired behavior. Therefore, we propose a novel training paradigm that reimagines contrastive decoding not as an inference patch, but as a self improving supervisory engine. Our approach tightly couples contrastive decoding, which serves as a dynamic reference for preference learning, with group relative policy optimization (GRPO) in a closed loop [DeepSeek-Math Blog, 2025 Preprint]. As the model improves, the pseudo ground truth generated by contrastive decoding becomes increasingly accurate and temporally faithful. The main contributions of our work can be summarized as follows:

- We propose an iterative training framework that fuses contrastive decoding with group relative policy optimization, creating a self-reinforcing loop that progressively internalizes temporal reasoning [Li et al., 2023; Chang et al., 2024; GRPO, 2025].
- We introduce TemporalQA, a benchmark evaluates a model’s sensitivity to time by asking fine grained questions about temporal understanding, enabling more accurate and realistic assessment of video temporal understanding compared to existing benchmarks[ReXTime (2024); V-STaR (2025); TemporalBench (2024)].
- We evaluate our method across four temporal reasoning benchmarks, including TemporalQA, and achieve state-of-the-art results.

2. Related Work

2.1. Temporal Reasoning in Video-LLMs

Temporal reasoning is crucial for understanding video content, as it enables models to grasp the order[5, 8–10], duration, and causality of events. Several studies have focused on enhancing temporal understanding in Video-LLMs.

Recent works divide the role of MLLMs in temporal reasoning into two main approaches: Facilitators[] and Executors[]. Facilitators, such as those used in EgoSchema and Di and Xie’s work, assist by transforming unstruc-

tured multimodal data into temporally structured representations. These methods primarily rely on static pre-trained priors and struggle with dynamic temporal awareness. On the other hand, Executors, represented by models like Grounded-VideoLLM, integrate temporal reasoning directly into the LLMs, processing video, text, and temporal cues together for end-to-end temporal reasoning. However, these approaches often face challenges with long-horizon temporal understanding and maintaining interpretability.

Our approach addresses these limitations by leveraging a novel framework that not only enhances temporal reasoning through contrastive decoding but also internalizes the temporal understanding via reinforcement learning, creating a self-reinforcing loop for continuous improvement.

2.2. Contrastive Decoding

Contrastive Decoding (CD) has gained attention for its potential to improve temporal reasoning by encouraging the model to compare temporally coherent versus disrupted outputs. While this approach has been used in both image and video domains[], it primarily serves as an inference-time correction. The Temporal Contrastive Decoding (TCD) method, for example, focuses on reducing hallucinations in video-event prediction by randomly dropping frames to generate perturbations. However, existing contrastive methods have a few key drawbacks: they double inference latency due to the need for two forward passes, show only modest performance gains, and fail to internalize the temporal cues they emphasize during inference.

In contrast, our work reimagines CD as a dynamic supervisory tool integrated within a reinforcement learning framework. By using CD as part of a reinforcement learning loop, we not only enhance the model’s output at inference but also embed temporal reasoning within the model’s parameters, enabling lasting improvements in temporal coherence.

2.3. Reinforcement Learning

Reinforcement learning (RL) has been increasingly applied to refine model behaviors through iterative feedback. In the context of Video-LLMs, RL is a powerful tool for improving model capabilities by using rewards based on task performance to guide learning. However, traditional RL methods are often limited by sparse or ambiguous feedback.

Our work innovates by introducing a self-reinforcing loop that couples Contrastive Decoding (CD) with Group Relative Policy Optimization (GRPO) in an RL framework. This loop allows the model to iteratively refine its temporal reasoning abilities by using contrastive signals as dynamic rewards, thus promoting long-term improvements without requiring extensive human annotations. Through this mechanism, our model enhances its temporal sensitivity, outperforming previous methods in temporal reasoning tasks.



Figure 1. Illustrative comparison of question design in existing benchmarks vs. TemporalQA.

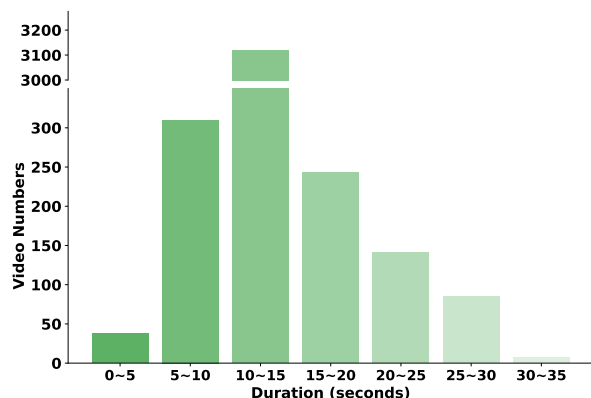


Figure 2. Distribution of video durations in the TemporalQA dataset.

Table 2. Statistics of the TemporalQA dataset. Here A.N.C. typically stands for “Average Number of Characters per QA annotation”.

Video		Annotation	
Total Length	Total Frames	Total samples	A.N.C
12.45h	1,222,547	3943	363.40

3. Dataset

Existing video question answering benchmarks often conflate temporal reasoning with general visual description or action recognition. Many rely on open-ended prompts (e.g. “Describe what happens in the video”), which do not explicitly probe the sensitivity of a model to time. As a result, poor performance on such tasks may reflect ambiguous task formulation rather than a true deficiency in temporal understanding.

To address this, we introduce TemporalQA, a benchmark specifically designed to assess the ability to direct temporal reasoning. Rather than soliciting free-form summaries, TemporalQA consists exclusively of closed-form questions that require precise judgments about event order, sequence continuity, and temporal polarity—e.g., “What occurs immediately after X?”, “Does A happen before or after B?”, “In which direction does the object move?”.

By framing evaluation around targeted questions, TemporalQA isolates the model’s ability to reason over time from confounding factors such as language modeling fluency or scene grounding accuracy. This makes it more discriminative, reliable, and aligned with real-world applications where temporal precision is critical.

4. Method

Our approach introduces a unified framework that synergistically combines video-based contrastive decoding (Video-CD) with a reinforcement learning (RL) feedback loop, enabling iterative refinement of temporal reasoning.

4.1. Video-based Contrastive Decoding

To strengthen the model’s capacity for temporal reasoning, we introduce a video-based contrastive decoding (Video-CD) scheme that operates during inference without modifying model parameters. The core idea is to explicitly encourage responses that are sensitive to the correct chronological order of events, by contrasting the model’s behavior under temporally coherent and temporally disrupted inputs.

For a given video-question pair (V, q) , we construct a temporally perturbed counterpart V_{rev} by reversing the sequence of uniformly sampled frames from V . This operation preserves all spatial and semantic content but inverts

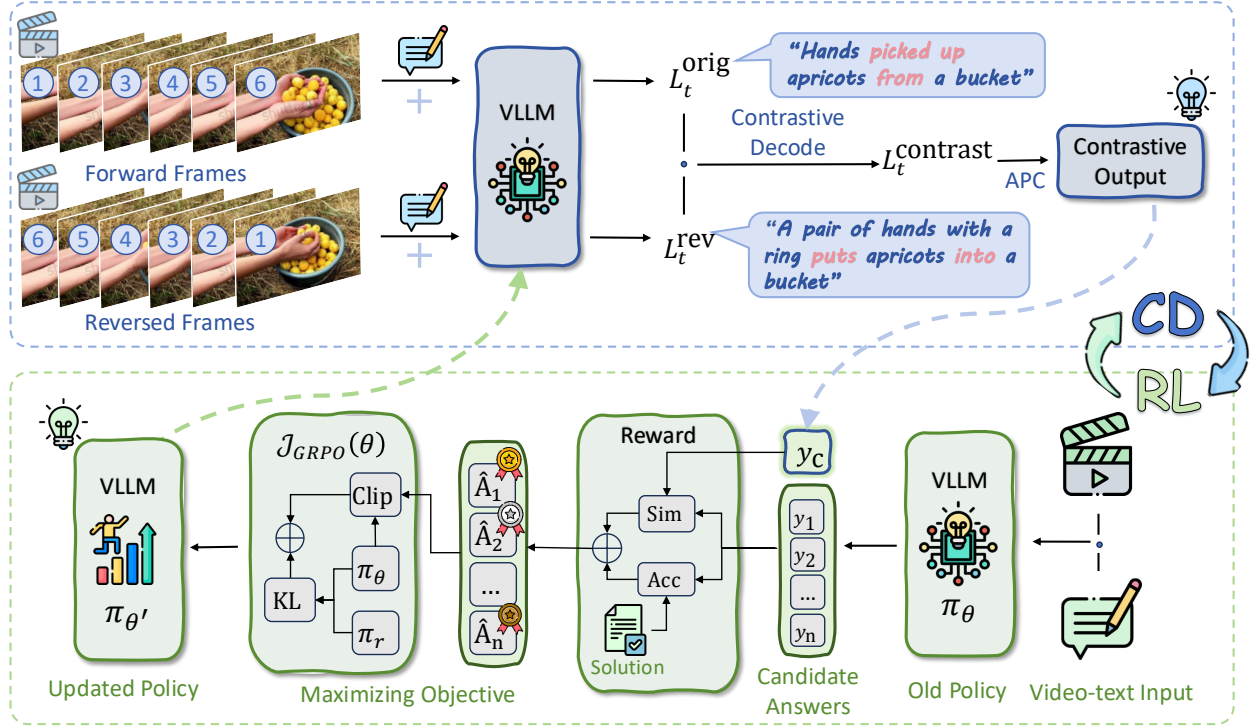


Figure 3. Overall architecture of the proposed RLCD framework.

the direction of action flow, thereby removing genuine sequential and causal cues while retaining static visual features.

During autoregressive generation, both V and V_{rev} are processed independently to obtain two sets of next-token logits:

$$L_t^{orig} = \text{logit}_\theta(y_t | V, q, y_{<t}) \quad (1)$$

$$L_t^{rev} = \text{logit}_\theta(y_t | V_{rev}, q, y_{<t}) \quad (2)$$

where y_t is the token to be generated and $y_{<t}$ denotes the previously generated sequence.

We then compute a contrastive logit distribution that amplifies tokens whose likelihood depends on correct event ordering:

$$L_t^{contrast} = (1 + \alpha)L_t^{orig} - \alpha L_t^{rev} \quad (3)$$

where $\alpha > 0$ is a hyperparameter.

Tokens that are highly probable under the original video but suppressed under the reversed video receive boosted scores, whereas tokens that remain consistently probable across both conditions (indicating they rely on static or time-agnostic cues) are attenuated.

To ensure robustness and prevent low-confidence tokens from dominating the contrastive signal, we apply an Adaptive Plausibility Constraint (APC). Specifically, we define a

dynamic cutoff threshold:

$$cutoff = \beta * \max(L_t^{orig}) \quad (4)$$

where $\beta \in (0, 1)$ controls the tolerance for plausible candidates. Only tokens satisfying $L_t^{orig} \geq cutoff$ are retained in $L_t^{contrast}$; all others are masked by setting their logits to $-\infty$.

4.2. Reinforcement Learning

While contrastive decoding improves inference-time responses, it does not update the model itself. To internalize temporal reasoning capabilities, we embed the contrastive output into an RL fine-tuning framework based on GRPO. For each training instance, we generate $y_{contrast}$ using the current model and sample multiple candidate answers $\{y_i\}$ via standard decoding. Each candidate is assigned a composite reward:

$$R(y_i) = \gamma \cdot \text{Acc}(y_i, \text{solution}) + \mu \cdot \text{Sim}(y_i, y_{contrast}) \quad (5)$$

where $\text{Acc}(\cdot, \cdot)$ reflects factual correctness with respect to ground-truth answers, and $\text{Sim}(\cdot, \cdot)$ measures alignment with the contrastive response—thereby rewarding outputs that preserve the temporal logic captured by $y_{contrast}$.

Following GRPO, the policy is updated by maximizing a clipped surrogate objective that balances reward improve-

ment and policy stability:

$$\begin{aligned} \mathcal{J}_{GRPO}(\theta) = & \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \\ & \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \right. \right. \\ & \left. \left. \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})}, 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_{i,t} \right] \right. \\ & \left. - \beta \mathbb{D}_{KL}[\pi_{\theta} || \pi_{ref}] \right\} \end{aligned} \quad (6)$$

where:

- ε and β are hyper-parameters
- $\hat{A}_{i,t}$ is the advantage calculated based on relative rewards of the outputs inside each group only.
- π_{ref} is a reference policy (e.g., the initial model) used to prevent excessive deviation.

Crucially, after this GRPO update, we obtain an improved policy $\pi_{\theta'}$. In the next iteration, we regenerate $y_{contrast}$ using $\pi_{\theta'}$. Because $\pi_{\theta'}$ has better temporal understanding (thanks to the previous RL update), the new $y_{contrast}$ becomes a more accurate and nuanced pseudo-ground truth, especially for questions requiring multi-step event reasoning.

This yields a self-amplifying loop:

$$\begin{array}{c} \pi_{\theta} \xrightarrow{\text{Contrastive Decoding}} y_{contrast} \\ \xrightarrow{\text{GRPO}} \pi_{\theta'} \xrightarrow{\text{Contrastive Decoding}} y'_{contrast} \xrightarrow{\text{GRPO}} \dots \end{array} \quad (7)$$

Each cycle refines the model’s internal representation of temporal causality and sequence dependence, gradually shifting its inductive bias toward time-aware reasoning without requiring additional human annotations.

5. Experiment

5.1. Datasets & Evaluation Metrics

RTime-QA is a benchmark of 822 human-annotated video-question pairs designed to evaluate atomic temporal event understanding, specifically the ability to distinguish forward versus reverse event progressions. Each test item is built upon a video pair depicting the same physical scene and objects but playing in opposite temporal directions, such that static frames are nearly identical and only the temporal dynamics differ. In our work, we convert the original multiple-choice format into open-ended QA, prompting models to generate a free-form description of the event in the given video rather than selecting between pre-defined options, thereby requiring robust reasoning over fine-grained temporal directionality.

TempCompass is a benchmark designed to assess fine-grained temporal understanding in Video LLMs across five

aspects (e.g., action, speed, direction, event order, and attribute change) and four task formats. Among these, we specifically use the **Caption Generation** task because it is the most challenging. This task requires models to first select the correct temporal information from multiple distractors and then generate a concise and faithful caption, thereby serving as a stringent test of both temporal reasoning and generation fidelity.

MVBench is a comprehensive benchmark designed for evaluating the temporal understanding capabilities of multimodal large language models on video inputs. It comprises 20 challenging tasks that cannot be solved using a single frame and span a wide spectrum from perception to high level cognition including action sequence reasoning, action prediction, object interaction, moving direction estimation, scene transition recognition, and counterfactual inference. This design ensures temporal sensitivity across diverse video scenarios and evaluation fairness without reliance on labor-intensive manual labeling.

5.2. Implementation Details

In contrastive decoding, we set the hyperparameters to $\alpha = 1.0$ and $\beta = 0.1$. We evaluate all models on four video QA benchmarks: RTime-QA, TempCompass, MVBench, and TemporalQA. For each model’s response, we employ GPT-3.5-turbo to assign scores following a standardized rubric to ensure consistent and objective evaluation. On RTimeQA, the reported strict Acc requires a model to correctly answer both questions in a paired sample (i.e., the original video and the reversed video). Only when both answers are correct is the pair counted as a true; otherwise, it is considered a failure. This metric emphasizes robust temporal reasoning and sensitivity to frame ordering.

5.3. Case Study

5.4. Ablation Study

To investigate the contribution of each component in our proposed RLCD framework, we conduct a comprehensive ablation study across four benchmark datasets: RTimeQA, TempCompass (Caption Generation), MVBench, and TemporalQA. We compare our full model (RLCD) with two ablated variants: (1) w/o all, which removes all key components introduced by RLCD (i.e., baseline without contrastive learning or temporal alignment modules); and (2) w/o ContrastSim, which retains other RLCD components but removes the ContrastSim module.

As shown in Table 4, removing all components (w/o all) leads to substantial performance degradation across all datasets, confirming that the architectural and training innovations in RLCD are essential for strong performance. For instance, on RTimeQA, accuracy drops from 85.89 to 68.49, and on TemporalQA, score decreases from 4.48 to 3.64, demonstrating the critical role of our design choices.

Table 3. Main experimental results on four temporal video QA benchmarks: RTimeQA, TempCompass, MVBench, and TemporalQA. Our method, RLCD, achieves state-of-the-art performance across all datasets in both accuracy and overall score metrics. Best results are highlighted in bold.

Method	Base Model	RTimeQA			TempCompass (Caption Generation)		MVBench		TemporalQA	
		Accuracy	Strict Acc	Score	Accuracy	Score	Accuracy	Score	Accuracy	Score
LLaVA-NeXT-Video	Vicuna-7B	69.10	51.34	3.52	57.04	3.22	45.66	2.83	58.26	3.36
MiniCPM-V-2.6	Qwen2-7B	60.83	37.71	3.30	59.73	3.61	47.18	3.00	60.21	3.41
VideoLLaVA	Vicuna-7B	58.03	38.20	3.26	45.76	2.66	53.34	2.85	52.75	3.11
LLaVA1.5	Vicuna-7B	53.89	33.58	2.90	48.55	3.27	38.57	2.72	61.05	3.63
VideoChat2	Mistral-7B	72.38	54.26	3.63	63.62	3.52	55.36	3.36	70.32	3.84
InternVL2	InternLM2.5-7B	67.64	48.42	3.61	58.73	3.23	55.40	3.46	63.55	3.66
LLaVA-OneVision	Qwen2-7B	55.72	39.42	2.78	63.32	3.49	61.23	3.28	69.15	3.90
LongVU	Qwen2-7B	–	–		53.34	3.13	64.78	3.63	62.28	3.51
RLCD (Ours)	Qwen2.5-VL-7B	85.89	75.43	4.41	67.66	3.86	65.41	3.83	85.68	4.48

Table 4. Ablation study of RLCD on four temporal video QA benchmarks. Removing the ContrastSim component (w/o ContrastSim) leads to significant performance drops, especially on TemporalQA (−5.38 in Accuracy), demonstrating its critical role. The “w/o all” variant removes all proposed components and serves as a baseline. Best results are in bold.

model	RTimeQA			TempCompass (Caption Generation)		MVBench		TemporalQA	
	Accuracy	Strict Acc	Score	Accuracy	Score	Accuracy	Score	Accuracy	Score
w/o all	68.49	50.36	3.64	62.13	3.61	62.01	3.69	64.87	3.64
w/o ContrastSim	77.12	62.29	3.95	64.67	3.74	63.59	3.75	80.30	4.27
RLCD (Ours)	85.89	75.43	4.41	67.66	3.86	65.41	3.83	85.68	4.48

When only the ContrastSim module is removed (w/o ContrastSim), performance still degrades significantly compared to the full model, albeit less severely than removing all components. On RTimeQA, Accuracy drops from 85.89 to 77.12, and on TemporalQA, Score falls from 4.48 to 4.27. This indicates that ContrastSim plays a pivotal role in improving temporal reasoning and captioning coherence particularly evident in TempCompass, where the score drops from 3.86 to 3.74 despite relatively high Accuracy (67.66 → 64.67). The consistent gap across metrics and datasets suggests that ContrastSim effectively aligns visual-temporal representations, thereby boosting both precision and semantic quality.

In summary, our ablation study validates that each component in RLCD contributes meaningfully to its superior performance, with ContrastSim being especially crucial for achieving state-of-the-art results in temporal video understanding tasks.

6. Introduction

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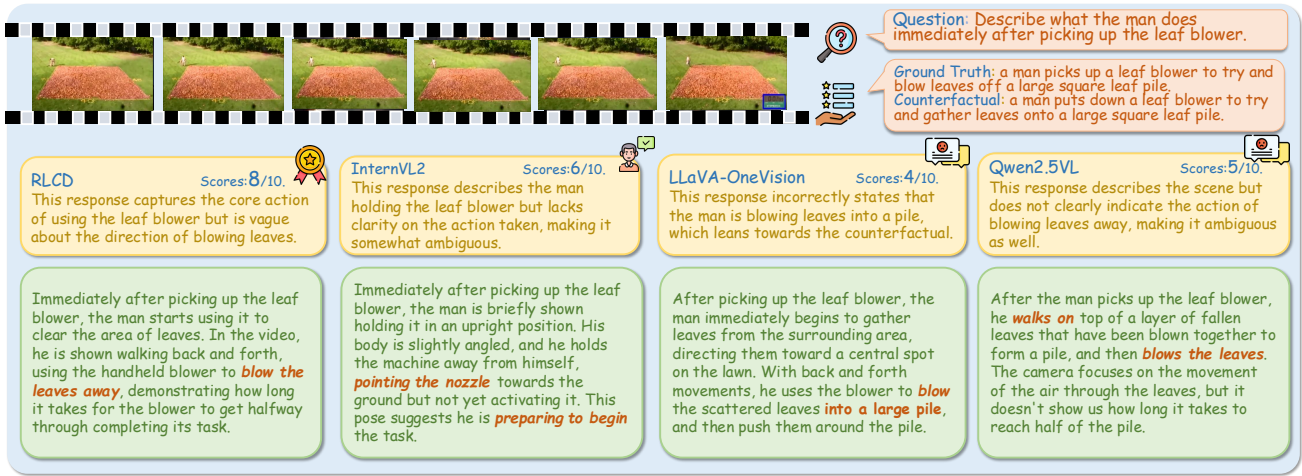


Figure 4. A case study on the TemporalQA dataset, comparing the outputs of four models on a representative time-sensitive QA example. This example illustrates differences in temporal expression recognition and reasoning abilities across models.

length of 10 pages. **There will be no extra page charges for CVPR 2026.**

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435	Blind review means that you do not use the words “my”	483	a key problem on the Apollo lander, and you believe that	483
436	or “our” when citing previous work. That is all. (But see	484	the 1970 audience would like to hear about your solution.	484
437	below for tech reports.)	485	The work is a development of your celebrated 1968 paper	485
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439	not mean that you are Lucy Smith; it says that you are build-	487	ing on her work. If you are Smith and Jones, do not say “as	487
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441	we show in [7]”, say “as Smith and Jones show in [7]” and	489	makes us a wow at parties”, by Zeus <i>et al.</i>	489
442	at the end of the paper, include reference 7 as you would	490	You can handle this paper like any other. Do not write	490
443	any other cited work.	491	“We show how to improve our previous work [Anonymous,	491
444	An example of a bad paper just asking to be rejected:	492	1968]. This time we tested the algorithm on a lunar lander	492
445	An analysis of the frobnicable foo filter.	493	[name of lander removed for blind review]”. That would be	493
446	In this paper, we present a performance analysis	494	silly, and would immediately identify the authors. Instead	494
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459	[1] Smith, L and Jones, C. “The frobnicable	507	As you can see, the above text follows standard scientific	507
460	foo filter, a fundamental contribution to human	508	convention, reads better than the first version, and does not	508
461	knowledge”. Nature 381(12), 1-213.	509	explicitly name you as the authors. A reviewer might think	509
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463	the same time that covers similar or overlapping material,	511	<i>al.</i> , but cannot make any decision based on that guess. He or	511
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466	related work. In such cases, include the anonymized paral-	514	FAQ	514
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468	[1] Authors. “The frobnicable foo filter”, F&G	516	A: No. Leave them for the final copy.	516
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470	tal material fg324.pdf.	518	A: To conform with the double-blind review policy, you	518
471	Finally, you may feel you need to tell the reader that	519	can report results of other challenge participants together	519
472	more details can be found elsewhere and refer them to a	520	with your results in your paper. However, for your results,	520
473	technical report. For conference submissions, the paper	521	you should not identify yourself and should not mention	521
474	must stand on its own, and not <i>require</i> the reviewer to go	522	your participation in the challenge. Instead, present your	522
475	to a tech report for further details. Thus, you may say	523	results referring to the method proposed in your paper and	523
476	in the body of the paper “further details may be found	524	draw conclusions based on the experimental comparison	524
477	in [7]”. Then submit the tech report as supplemental ma-	525	with other results.	525
478	terial. Again, do not assume that the reviewers will read	526	6.8. Miscellaneous	526
479	this material.	527	Compare the following:	527
480	Sometimes your paper is about a problem that you tested	528	$\$conf_a\$$ $conf_a$	528
481	using a tool that is widely known to be restricted to a single	529	$\$\mathit{conf}_a\$$ $conf_a$	529
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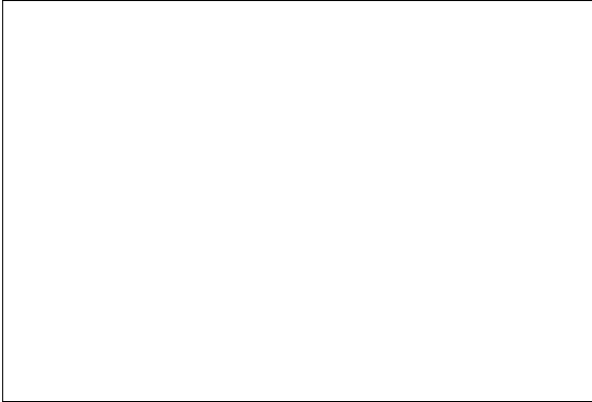


Figure 5. Example of caption. It is set in Roman so that mathematics (always set in Roman: $B \sin A = A \sin B$) may be included without an ugly clash.

The space after *e.g.*, meaning “for example”, should not be a sentence-ending space. So *e.g.* is correct, *e.g.* is not. The provided `\eg` macro takes care of this.

When citing a multi-author paper, you may save space by using “et alia”, shortened to “*et al.*” (not “*et. al.*” as “*et*” is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al.* However, use it only when there are three or more authors. Thus, the following is correct: “Frobnication has been trendy lately. It was introduced by Alpher [1], and subsequently developed by Alpher and Fotheringham-Smythe [2], and Alpher *et al.* [4].”

This is incorrect: “... subsequently developed by Alpher *et al.* [2] ...” because reference [2] has only two authors.

7. Formatting your paper

All text must be in two-column format. The total allowable size of the text area is $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. The columns should be $3\frac{1}{4}$ inches (8.25 cm) wide, with a $\frac{5}{16}$ inch (0.8 cm) space between them. The main title (on the first page) should begin 1 inch (2.54 cm) from the top edge of the page. The second and following pages should begin 1 inch (2.54 cm) from the top edge. On all pages, the bottom margin should be $1\frac{1}{8}$ inches (2.86 cm) from the bottom edge of the page for 8.5×11 -inch paper; for A4 paper, approximately $1\frac{5}{8}$ inches (4.13 cm) from the bottom edge of the page.

7.1. Margins and page numbering

All printed material, including text, illustrations, and charts, must be kept within a print area $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. Page numbers should be in the footer, centered, and $\frac{3}{4}$ inches from the bottom of the page. The review version should have page numbers, yet the final version submitted as camera ready should not

show any page numbers. The $\text{\LaTeX}$ template takes care of this when used properly.

7.2. Type style and fonts

Wherever Times is specified, Times Roman may also be used. If neither is available on your word processor, please use the font closest in appearance to Times to which you have access.

MAIN TITLE. Center the title $1\frac{3}{8}$ inches (3.49 cm) from the top edge of the first page. The title should be in Times 14-point, boldface type. Capitalize the first letter of nouns, pronouns, verbs, adjectives, and adverbs; do not capitalize articles, coordinate conjunctions, or prepositions (unless the title begins with such a word). Leave two blank lines after the title.

AUTHOR NAME(s) and AFFILIATION(s) are to be centered beneath the title and printed in Times 12-point, non-boldface type. This information is to be followed by two blank lines.

The ABSTRACT and MAIN TEXT are to be in a two-column format.

MAIN TEXT. Type main text in 10-point Times, single-spaced. Do NOT use double-spacing. All paragraphs should be indented 1 pica (approx. $\frac{1}{6}$ inch or 0.422 cm). Make sure your text is fully justified—that is, flush left and flush right. Please do not place any additional blank lines between paragraphs.

The captions of the figures and tables should be in 9-point Roman type as in Figs. 5 and 6. Short captions should be centered. Table captions should be above tables, while figure captions should be below figures.

Callouts should be 9-point Helvetica, non-boldface type. Initially capitalize only the first word of section titles and first-, second-, and third-order headings.

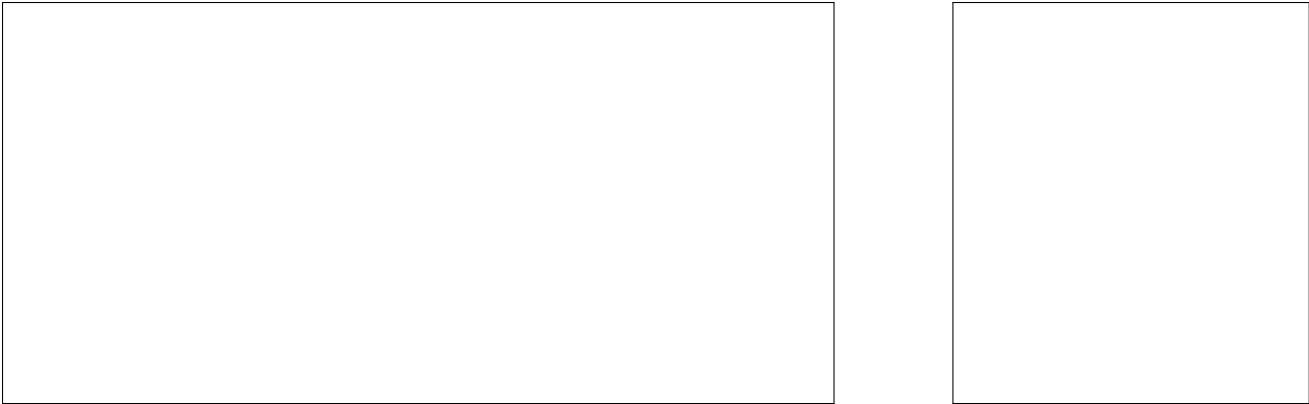
FIRST-ORDER HEADINGS. (For example, **1. Introduction**) should be Times 12-point boldface, initially capitalized, flush left, with one blank line before and one blank line after.

SECOND-ORDER HEADINGS. (For example, **1.1. Database elements**) should be Times 11-point boldface, initially capitalized, flush left, with one blank line before and one after. If you require a third-order heading (we discourage it), use 10-point Times, boldface, initially capitalized, flush left, preceded by one blank line, followed by a period, and your text on the same line.

7.3. Footnotes

Please use the footnotes¹ sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the

¹This is what a footnote looks like. It often distracts the reader from the main flow of the argument.



(a) An example of a subfigure. (b) Another example of a subfigure.

Figure 6. Example of a short caption, which should be centered.

bottom of the column on the page on which it is referenced.
Use Times 8-point type, single-spaced.

7.4. Cross-references

For the benefit of author(s) and readers, please use the

`\cref{...}`

command for cross-referencing to figures, tables, equations,
or sections. This will automatically insert the appropriate
label alongside the cross-reference as in this example:

To see how our method outperforms previous
work, see Fig. 5 and Tab. 5. It is also possible
to refer to multiple targets as once, *e.g.* to Figs. 5
and 6a. You may also return to Sec. 7 or look at
Eq. (9).

If you do not wish to abbreviate the label, for example, at
the beginning of the sentence, you can use

`\Cref{...}`

command. Here is an example:

Figure 5 is also quite important.

7.5. References

List and number all bibliographical references in 9-point
Times, single-spaced, at the end of your paper. When ref-
erenced in the text, include the citation number in square
brackets, for example [6]. Where appropriate, include page
numbers and the name(s) of editors of referenced books.
When citing multiple papers at once, make sure that you
cite them in numerical order such as this [1–3, 6, 7]. If you
use the template as advised, this will be taken care of auto-
matically.

Table 5. Results. Ours is better.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one’s heart Frob

7.6. Illustrations, graphs, and photographs

All graphics should be centered. In $\LaTeX$, avoid using the
`center` environment for this purpose, as this adds poten-
tially unwanted whitespace. Instead, use

`\centering`

at the beginning of your figure. Please ensure that any point
you wish to make is resolvable in a printed copy of the pa-
per. Resize fonts in figures to match the font in the body
text, and choose line widths that render effectively in print.
Readers (and reviewers), even of an electronic copy, may
choose to print your paper in order to read it. You cannot
insist that they do otherwise and, therefore, must not assume
that they can zoom in to see tiny details on a graphic.

When placing figures in $\LaTeX$, it is almost always best to
use `\includegraphics`, and to specify the figure width
as a multiple of the line width as in the example below

```
\usepackage{graphicx} ...  
\includegraphics[width=0.8\linewidth]  
{myfile.pdf}
```

7.7. Color

Please refer to the author guidelines on the CVPR 2026 web
page for a discussion of the use of color in your document.

If you use color in your plots, please keep in mind that a
significant subset of reviewers and readers may have a color
vision deficiency; red-green blindness is the most frequent

kind. Hence, avoid relying only on color as the discriminative feature in plots (such as red vs. green lines), but add a second discriminative feature to ease disambiguation.

8. Final copy

You must include your signed IEEE copyright release form when you submit your finished paper. We MUST have this form before your paper can be published in the proceedings.

Please direct any questions to the production editor in charge of these proceedings at the IEEE Computer Society Press: <https://www.computer.org/about/contact>.

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